ABINAYA SENTHILNATH

Mail · Linkdin · Github · Portfolio

SUMMARY

An Automation and Robotics Engineer aiming to transition into an AI and Data Science role focused on Intelligent Robotics. I enjoy working on bringing intelligence to hardware, combining practical experience in autonomous robotic systems and IoT with machine learning and computer vision for real-world robotics applications

PROFESSIONAL EXPERIENCE

Embedded Developer **05/2025 - 06/2025**
MachDatum

• Developed and implemented an IoT-based Home Security Automation System using Arduino, MQTT, and PostgreSQL, enhancing system security and data management
• Designed functional product prototypes and contributed to system integration and testing, improving hardware-software compatibility
• Ensured 100% security in and out of the house, without any privacy invasions

EDUCATION

M.tech in Artificial Intelligence **July 2026 - April 2028**
Amrita school of computing, Amrita Vishwa vidyapeetham

B.tech in Automation and Robotics **September 2022 - April 2026**
Amrita school of Engineering, Amrita Vishwa vidyapeetham

BVM Global **July 2011 - March 2022**

PROJECTS

1. **Decision Support System for Mango Farming**

Developed and deployed an AI-based precision farming system incorporating Explainable AI for mango variety classification, disease localisation, fruit grading, and ripeness prediction, achieving >90% trust and accuracy among farmers. (*Paper published in Springer-IJCACI2025*)

2. **Neural Network-Based Acoustic Detection of Red Palm Weevil Infestation in Coconut Trees**

Developed an AI system for early detection of infestations using acoustic signals, a lightweight neural network trained on MFCC and spectral contrast features, achieved 86% accuracy.
(*Paper Published in IEEE- DELCON2025*)

3. **Autonomous Indoor Localisation & Guidance System**

Developed an algorithm for optimal placement of ArUco Tags in complex indoor environments. The algorithm validated with MATLAB, Webots, and hardware setups achieved localisation accuracy >90% across all cases.

SKILLS & LANGUAGES

Programming Languages: Python, C++
AI/ML Frameworks: TensorFlow, Keras, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, OpenCV
Simulation & Design Tools: ROS2, Webots, Fusion 360, AutoCAD, Adams, Arena, Ultimaker CURA
Microcontrollers: Arduino, ESP32, Raspberry Pi
Development Tools: VS Code, PyCharm, Jupyter, MATLAB, GitHub, LabVIEW, CODESYS
Office & Firmware: EV3 Mindstorms Firmware, MS Word, PowerPoint
General Skills: Technical Communication, Team Collaboration, Project Management,

English - Proficient, **Tamil** - Proficient, **Hindi** - Intermediate